Digital Game Design 🎮

Cadette Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors

Mechanics make feelings

Designers **engineer emotion** on purpose:

Mechanic	Feeling it manufactures
Countdown timer	tension
Growing combo counter	mastery
A near-miss	“one more try”

Dissect one you know 

Pick a game this room plays.

- What's its **core mechanic**?
- What **feeling** does that mechanic manufacture?

The ethics beat ⚖️

The same tools can **respect** players — or **hook** them unfairly.

Designing *for good* means fun that leaves players glad they played.

That’s a choice you make. Every time.

Design doc — with a curve 

One page, before any code:

1. One **core mechanic**
2. **Goal**
3. **Fail state**
4. **Intended feeling**
5. A sketched **difficulty curve**

The difficulty curve 📈

How does minute three get harder than minute one?

- Faster enemies?
- Shorter timer?
- More obstacles?

Scope rule: one mechanic executed well beats four executed poorly.

Build the prototype 💻

1. **Core loop** — player, objective, fail state
2. **Score** + win/lose conditions
3. **The difficulty curve, implemented** — a variable that ramps speed, spawn rate, or time pressure as score climbs
4. **Game feel pass** — sound effects, a screen shake or flash
5. Stretch: title screen + “play again” flow

The game feel pass ✨

Tiny touches. Huge difference.

This is the minute your prototype starts feeling like a real game.

Structured playtest 🔬

Not a swap. A protocol.

Half one — observe silently, 2 minutes: Where do they struggle? Smile? Quit?

Half two — interview, 3 questions: 1. What was the goal? 2. What frustrated you? 3. What would you change?

Watch, don’t explain 🤐

You will *ache* to explain your game mid-test.
Don’t.

What testers do outranks what they say.

Iterate from the data 🔧

Implement **one change** from your protocol sheet.

Closing: name the observation that surprised you most.

That surprise is why studios playtest.

Robot moment 

Every season a brand-new competition game drops.

Our first job is exactly this analysis — mechanics, scoring, fail states — before we design a robot to win it.

Same analysis. 120-pound game piece.

What you accomplished ✅

- Explained how a mechanic creates a specific player feeling
- Wrote a design doc with a core mechanic, fail state, and difficulty curve
- Built a prototype that implements that curve
- Ran a structured playtest and iterated on observed data

Level up 🚀

- **arcade.makecode.com** — add your title screen and “play again” flow
- Run your **playtest protocol** on a game someone else made
- Pick any game you play: name its core mechanic and the feeling it engineers

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